Data-Driven Pricing Intelligence for WOW! Agents

HDB Resale Price Market Analysis & Predictive Modeling

The Model Citizens·25 April 2025

Meet the architecture behind the WOW! Pricing Tool

Chun Wei

Product Owner & Lead

Core Tools

PPT ·Streamlit App

Keeping the team on track

Chauncey

Data Engineer & Analyst

Core Tools

Data Cleaning · EDA

Making the data behave

Irin

Visualisation Specialist

Core Tools

Tableau· EDA

Making the numbers look good

Shi Huan

ML Engineer

Core Tools

Linear Regression · Metrics

Building the brain behind the tool

From raw HDB datasets to a deployable pricing engine

01

Data Ingestion

Load & inspect HDB dataset.

167,371 rows-78 columns

02

Data Cleaning

Handle nulls, outliers, and feature engineering.

03

Exploratory Data Analysis

Identify price drivers via visualisations.

04

Modelling

Linear regression baseline,train/test splits, advanced metrics.

05

Deployment

Streamlit app tailored for agent use.

Refining the dataset for precision modeling

Data Pipeline Transformations

Missing Values	Mall Nearest Distance replaced with NA.Hawkers & Malls within 1km replaced with 0.
Outliers	Replaced using median values due to extreme numerical variances.
Engineering	Binned floor_area_sqft and cutoff_point. Generated new resale price target for test data.
Deduplication	Verified 100% unique entries across all records.

Cleaned Payload

150,634

Total Rows (Train)

16,737

Total Rows (Test)

78 → 68

Cleaned Columns

14

Primary Features Explored

The Analytical Lens

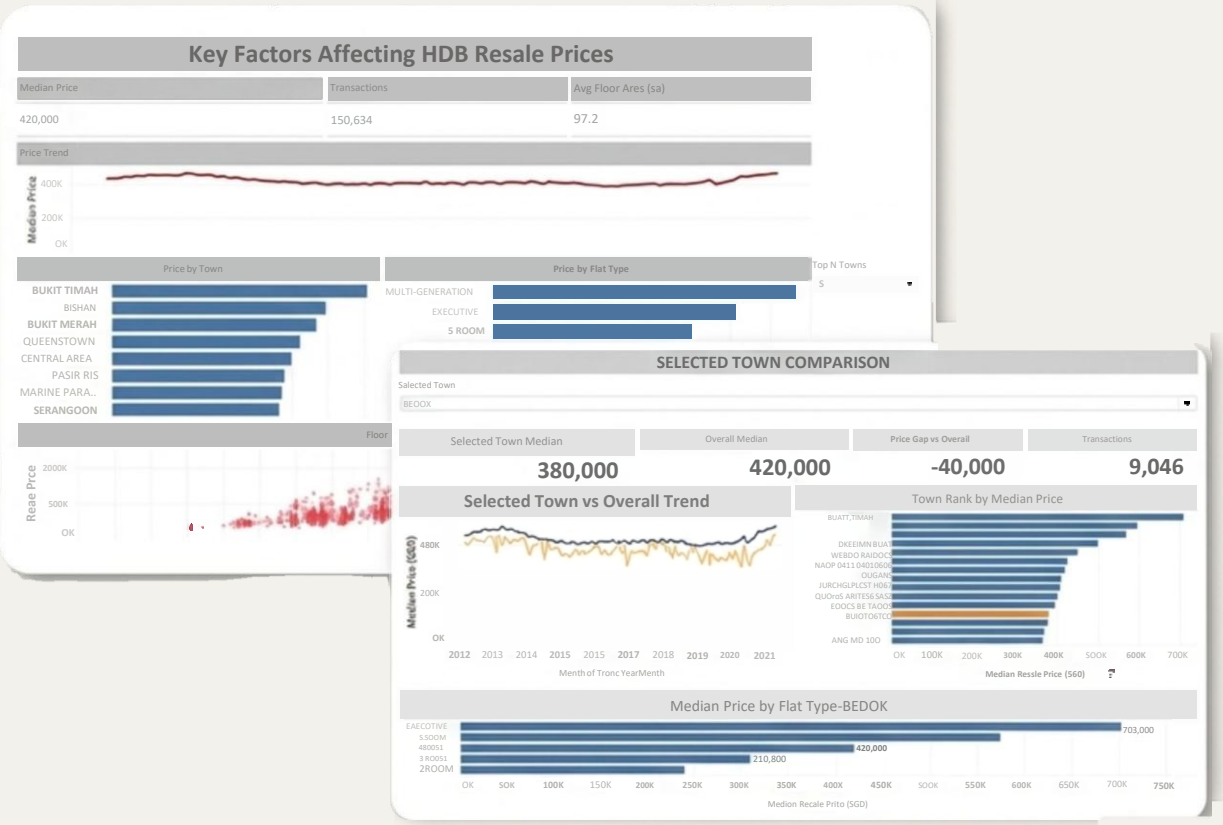
Location and flat type define the foundational resale value

Macro Drivers

EDA reveals that HDB prices are primarily anchored by Town/Location and Flat Type.

Floor area demonstrates a strong, undeniable positive relationship with the final resale price.

Local town pricing (e.g., Bedok) can deviate from overall medians while still mirroring broader macro market trend lines over time.



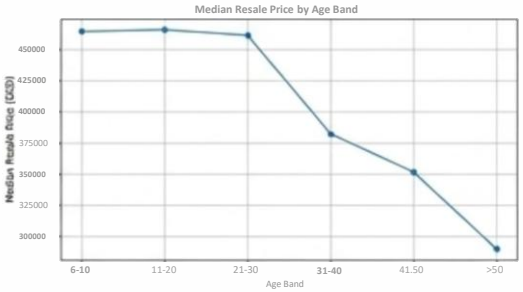
The Analytical Lens

Age, altitude, and amenities act as secondary price multipliers

Storey Level Impact

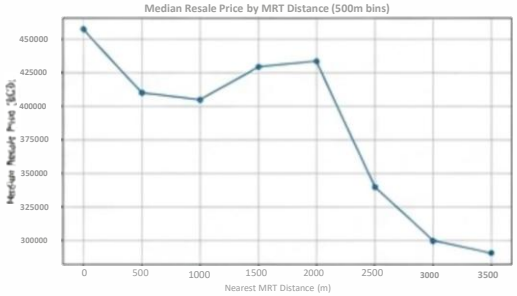
Higher storeys consistently command higher prices.

HDB Age & Lease Decay



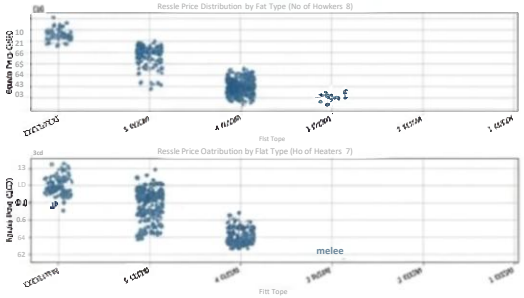
Noticeable value drop as flats age past 30 years and consume more lease.

MRT Accessibility



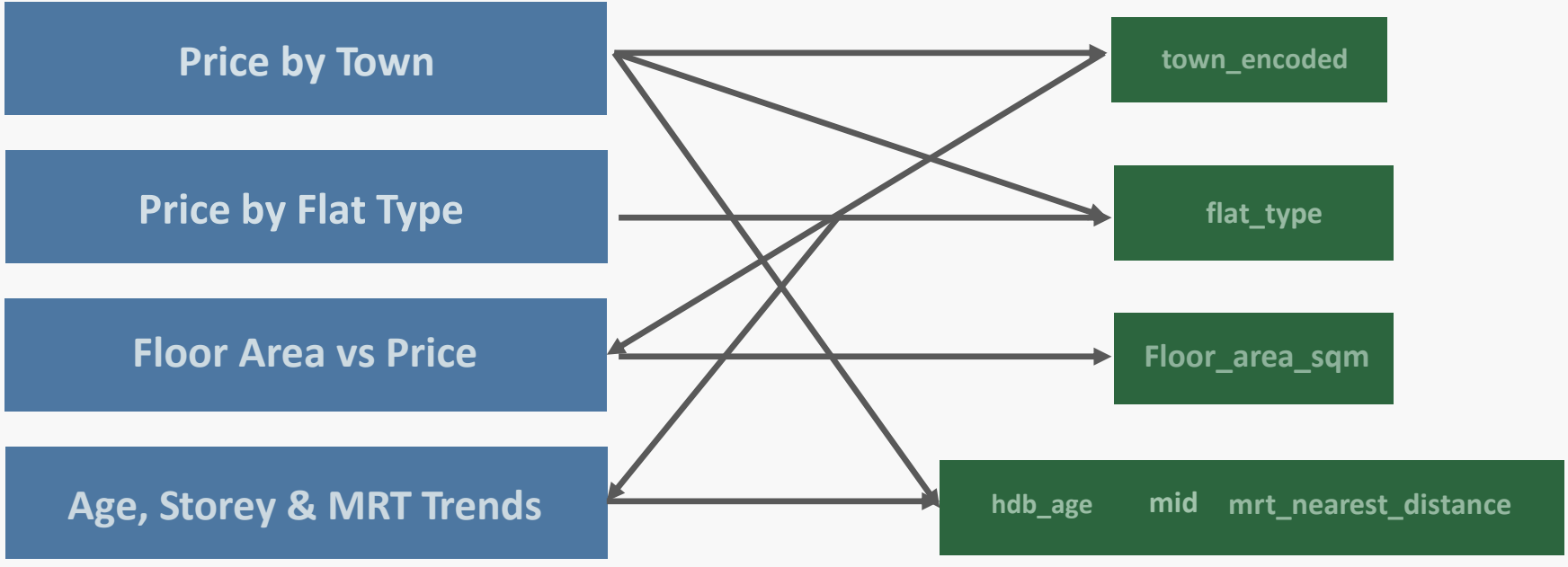
Steep price drop-offs for flats located further than 2,000m from the nearest transit hub.

Amenity Distortions



Mall and hawker counts influence price, but the signal requires careful interpretation due to flat-mix distortions in highly dense areas.

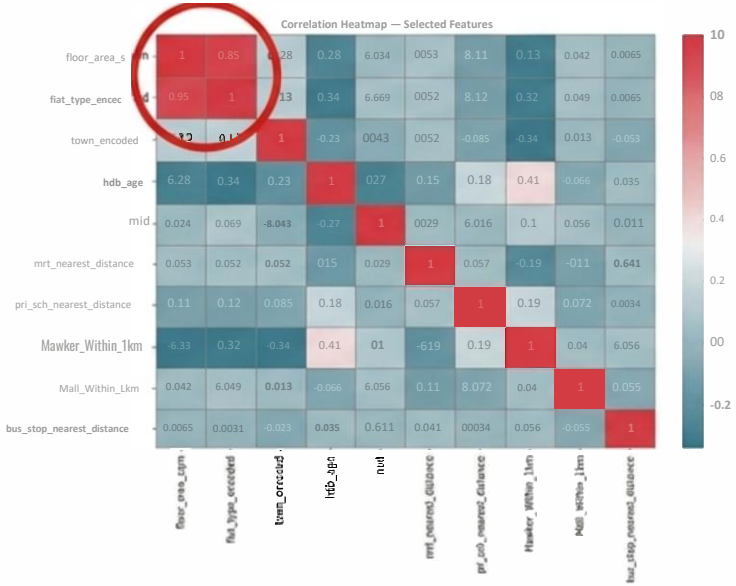
Codifying market intuition into machine-readable signals



By explicitly linking visual patterns to specific columns, we ensure the model learns from genuine market logic rather than random noise.

Resolving signal conflict to stabilize the algorithm

The Multicollinearity Conflict



Strong overlap (≥0.80) creates severe multicollinearity.

Initial State VIF

	Feature	VIF
0	const	55.579842
1	floor_area_sqm	10.568375
2	flat_type_encoded	10.839690
3	town_encoded	1.148903
4	hdb_age	1.510445
5	mid	1.155671
6	mrt_nearest_distance	1.072444
7	pri_sch_nearest_distance	1.069547
8	Hawker_Within_1km	1.510473
9	Mall_Within_1km	1.032356
10	bus_stop_nearest_distance	1.011371

Final State VIF

	Feature	VIF
0	const	50.004031
1	flat_type_encoded	1.189519
2	town_encoded	1.148437
3	hdb_age	1.483547
4	mid	1.149098
5	mrt_nearest_distance	1.072255
6	pri_sch_nearest_distance	1.069546
7	Hawker_Within_1km	1.492446
8	Mall_Within_1km	1.032319
9	bus_stop_nearest_distance	1.011208

VIF: Variance Inflation Factor

*Decision: We removed **floor_area_sqm** and kept **flat_type_encoded**, bringing all VIF values below 5 and producing a* acceptable, low correlation *feature set for the final predictive model run while preserving essential structural categorization.*

Optimizing the feature payload for maximum accuracy

4 Features	(Type,Town,Age,Storey)	Train RMSE: 95,616	Test RMSE: 95,725
5 Features	(+ MRT Distance)	Train RMSE: 92,109	Test RMSE: 92,216
6 Features	(+ Primary School)	Train RMSE: 91,322	Test RMSE: 91,474
7 Features	(+ Hawker Within 1km)	Train RMSE: 82,321	Test RMSE: 82,503
8 Features	(+ Mall Within 1km)	Train RMSE: 82,313	Test RMSE: 82,498
9 Features	**(+ Bus Stop Distance)**	**Train RMSE: 82,265**	**Test RMSE: 82,458 (Optimal)**

Takeaway: The 9-feature model delivered the lowest Train and Test RMSE with a minimal gap, confirming a well-fitted model with strong predictive accuracy

Why XGBoost was chosen as the underlying engine

Model Shootout

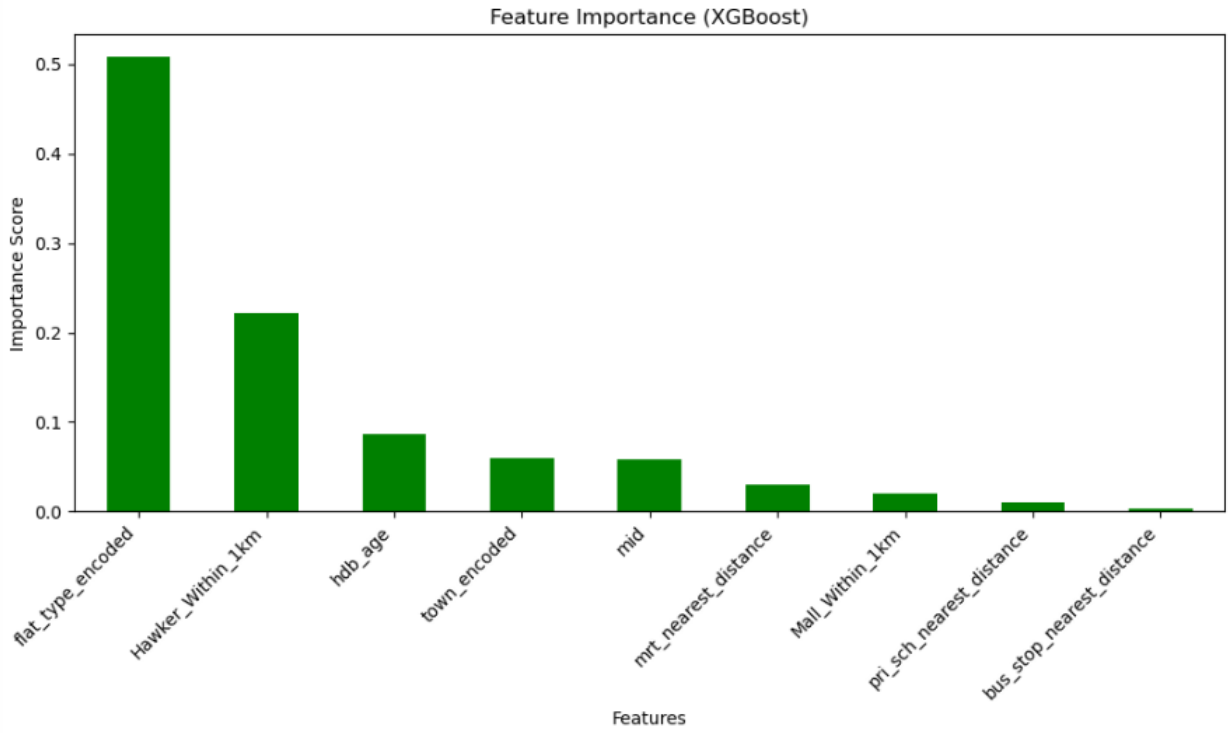
Linear Regression (Baseline)	Random Forest	XGBoost (Winner)
Train R2	Train R2	Train R2
0.67	0.96	0.92
Test RMSE	Test RMSE	Test RMSE
82,458	40,253	38,974
CV RMSE	CV RMSE	CV RMSE
82,271	40,324	39,254
Verdict	Verdict	Verdict
Underperformed. Fails to capture non-linear HDB market complexities.	Competitive, but validation performance is slightly less stable.	Strongest balance of accuracy and generalization.

Cross-Validation

Metric	Random Forest	XGBoost
Accuracy	0.883	0.899
Precision	0.884	0.899
Recall	0.884	0.899
F1 Score	0.884	0.899
Bias	0.384	0.399

XGBoost give us the best balance if accuracy, stability and generalization to be selected as the final model

Validating the machine's logic against market realities



0.9259

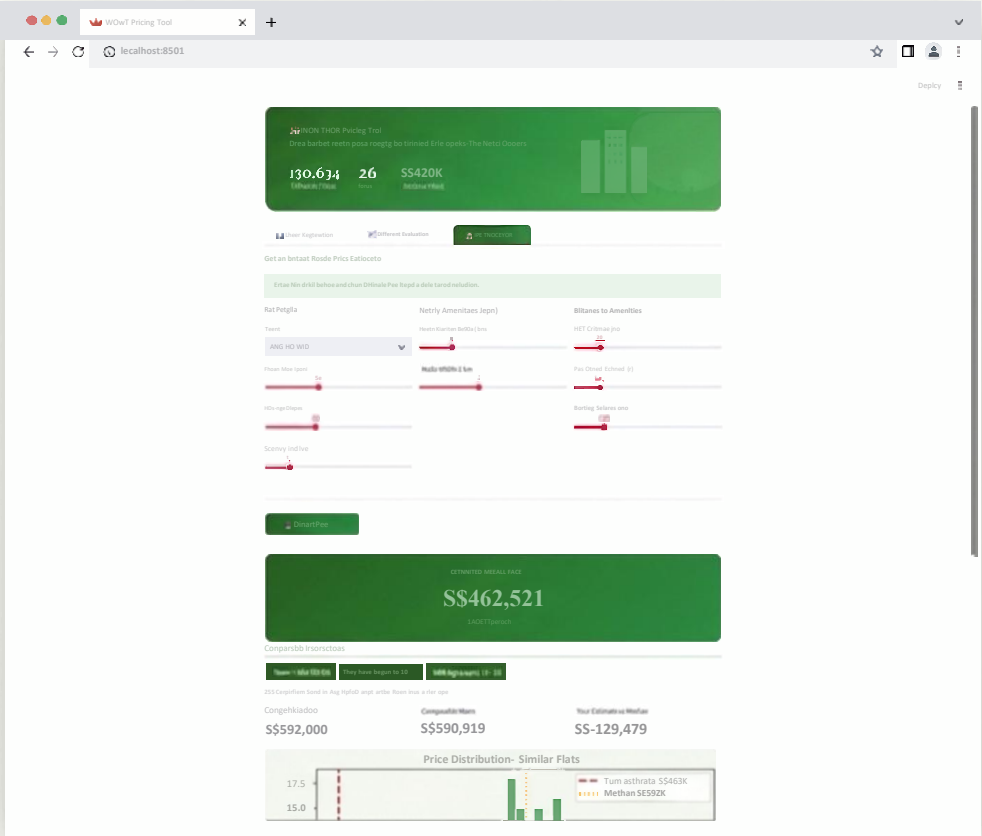
Test Adjusted R^2 (Variance efficiently explained)

$38,974

Test RMSE (Root mean square error)

Key Insight: The model does not rely on a single dominant variable. It successfully combines structural signals (space factor) and neighborhood constraints to estimate value. This confirms it is learning genuine market mechanics rather than capturing random noise.

The WOW! Pricing Tool: Instant data-backed valuations



1

1.Input Flat Details

Select Town, Floor Area, Age, and Mid-Storey level.

2

2.Define Environment

Toggle counts for nearby amenities (Hawkers/Malls) and exact distances to transport/schools.

3

3.Generate Intelligence

Get an instant baseline price estimate, benchmarked against real comparable transactions and price distribution charts.

Strategic Playbook for WOW! Agents

Prioritise Flat Type & Storey

Larger categories (Multi-Gen, Executive, 5 - Room) and higher storeys command heavy premiums. Guide buyers accordingly.

Town

Centrally located towns such as Bukit Timah or Bishan consistently command premiums above newer suburban estates such as Punggol or Sengkang.

Factor in Flat Age

Older flats face measurable lease decay penalties. Always benchmark against similar-age comparables when managing seller expectations.

Human + Machine

Use the WOW! tool for an instant, data-backed baseline, but manually refine it using agent judgment on micro-location, internal condition, and recent unrecorded transactions.

Sprint Retrospective & Next Priorities

✅ Wins

- Effective task distribution.
- Dividing workflows across cleaning, EDA, visualization, modeling and Streamlit deployment allowed rapid parallel progress.

⚠️ Challenges

- Aligning the human EDA story with the final ML feature set required multiple revisions (especially managing correlated competing features).
- A strong presentation requires an intentional bridge from EDA to ML.

➡️ Next Steps

- Implement stricter version control, lock in feature definitions earlier in the timeline, and optimize the handoff pipeline via dedicated Trello boards for better sprint workflow ownership.

Questions?

Data-Driven Pricing Intelligence for WOW! Agents

The Model Citizens

Thank you

The Model Citizens · Chun Wei · Chauncey · Shi Huan · Irin